```
/* ============================================================
   PL/SQL COMPLETE EXAMPLES
   Topics:
   1. IF THEN
   2. IF THEN ELSE
   3. IF THEN ELSIF
   4. FOR LOOP
   5. WHILE LOOP
   6. SIMPLE LOOP
   7. NESTED FOR LOOP
   8. PROCEDURE
   9. FUNCTION
   10. EXCEPTION
   ============================================================ */

SET SERVEROUTPUT ON;

/* ============================================================
   1. IF THEN Example
   ============================================================ */
```

```
SQL> DECLARE
    v_salary NUMBER := 800000;
BEGIN
    IF v_salary > 500000 THEN
        DBMS_OUTPUT.PUT_LINE('Salary is greater than 500,000');
    END IF;
END;
/
  2    3    4    5    6    7    8  Salary is greater than 500,000

PL/SQL procedure successfully completed.
```

```
/* ============================================================
   2. IF THEN ELSE Example
   ============================================================ */
```

```
SQL> DECLARE
    v_age NUMBER := 20;
BEGIN
    IF v_age >= 18 THEN
        DBMS_OUTPUT.PUT_LINE('User is adult');
    ELSE
        DBMS_OUTPUT.PUT_LINE('User is under age');
    END IF;
END;
/
  2    3    4    5    6    7    8    9   10  User is adult

PL/SQL procedure successfully completed.
```

```
/* ============================================================
   3. IF THEN ELSIF Example
   ============================================================ */
```

```
SQL> DECLARE
    v_mark NUMBER := 75;
BEGIN
    IF v_mark >= 80 THEN
        DBMS_OUTPUT.PUT_LINE('Grade A');
    ELSIF v_mark >= 60 THEN
        DBMS_OUTPUT.PUT_LINE('Grade B');
    ELSIF v_mark >= 40 THEN
        DBMS_OUTPUT.PUT_LINE('Grade C');
    ELSE
        DBMS_OUTPUT.PUT_LINE('Fail');
    END IF;
END;
/
  2    3    4    5    6    7    8    9   10   11   12   13   14  Grade B

PL/SQL procedure successfully completed.
```

```
/* ============================================================
   4. FOR LOOP Example
   ============================================================ */
```

```
SQL> BEGIN
    FOR i IN 1..5 LOOP
        DBMS_OUTPUT.PUT_LINE('FOR LOOP value: ' || i);
    END LOOP;
END;
/
  2    3    4    5    6  FOR LOOP value: 1
FOR LOOP value: 2
FOR LOOP value: 3
FOR LOOP value: 4
FOR LOOP value: 5

PL/SQL procedure successfully completed.
```

```
/* ============================================================
   5. WHILE LOOP Example
   ============================================================ */
```

```
SQL> DECLARE
    v_counter NUMBER := 1;
BEGIN
    WHILE v_counter <= 5 LOOP
        DBMS_OUTPUT.PUT_LINE('WHILE LOOP value: ' || v_counter);
        v_counter := v_counter + 1;
    END LOOP;
END;
/
  2    3    4    5    6    7    8    9  WHILE LOOP value: 1
WHILE LOOP value: 2
WHILE LOOP value: 3
WHILE LOOP value: 4
WHILE LOOP value: 5

PL/SQL procedure successfully completed.
```

```
/* ============================================================
   6. SIMPLE LOOP Example
   ============================================================ */
```

```
SQL> DECLARE
    v_counter NUMBER := 1;
BEGIN
    LOOP
        DBMS_OUTPUT.PUT_LINE('SIMPLE LOOP value: ' || v_counter);

        v_counter := v_counter + 1;

        EXIT WHEN v_counter > 5;
    END LOOP;
END;
/
  2    3    4    5    6    7    8    9   10   11   12  SIMPLE LOOP value: 1
SIMPLE LOOP value: 2
SIMPLE LOOP value: 3
SIMPLE LOOP value: 4
SIMPLE LOOP value: 5

PL/SQL procedure successfully completed.
```

```
/* ============================================================
   7. NESTED FOR LOOP Example
   Multiplication Table
   ============================================================ */
```

```
SQL> BEGIN
    FOR i IN 1..3 LOOP
        FOR j IN 1..5 LOOP
            DBMS_OUTPUT.PUT_LINE(i || ' x ' || j || ' = ' || (i * j));
        END LOOP;

        DBMS_OUTPUT.PUT_LINE('-------------------');
    END LOOP;
END;
/
  2    3    4    5    6    7    8    9   10  1 x 1 = 1
1 x 2 = 2
1 x 3 = 3
1 x 4 = 4
1 x 5 = 5
-------------------
2 x 1 = 2
2 x 2 = 4
2 x 3 = 6
2 x 4 = 8
2 x 5 = 10
-------------------
3 x 1 = 3
3 x 2 = 6
3 x 3 = 9
3 x 4 = 12
3 x 5 = 15
-------------------

PL/SQL procedure successfully completed.
```

```
/* ============================================================
   Create Sample Table
   ============================================================ */
DROP TABLE students;

CREATE TABLE students
(
    student_id NUMBER PRIMARY KEY,
    student_name VARCHAR2(100),
    mark NUMBER,
    grade VARCHAR2(10)
);

INSERT INTO students VALUES (1, 'Aung Aung', 85, NULL);
INSERT INTO students VALUES (2, 'Mg Mg', 70, NULL);
INSERT INTO students VALUES (3, 'Su Su', 45, NULL);
INSERT INTO students VALUES (4, 'Hla Hla', 30, NULL);

COMMIT;
```

```
/* ============================================================
   8. PROCEDURE Example
   Procedure to update student grade
   ============================================================ */
CREATE OR REPLACE PROCEDURE update_student_grade
(
    p_student_id IN NUMBER
)
AS
    v_mark students.mark%TYPE;
    v_grade VARCHAR2(10);
BEGIN
    SELECT mark
    INTO v_mark
    FROM students
    WHERE student_id = p_student_id;

    IF v_mark >= 80 THEN
        v_grade := 'A';
    ELSIF v_mark >= 60 THEN
        v_grade := 'B';
    ELSIF v_mark >= 40 THEN
        v_grade := 'C';
    ELSE
        v_grade := 'Fail';
    END IF;

    UPDATE students
    SET grade = v_grade
    WHERE student_id = p_student_id;

    COMMIT;

    DBMS_OUTPUT.PUT_LINE('Student grade updated successfully');
EXCEPTION
    WHEN NO_DATA_FOUND THEN
        DBMS_OUTPUT.PUT_LINE('Student not found');
    WHEN OTHERS THEN
        DBMS_OUTPUT.PUT_LINE('Error: ' || SQLERRM);
END;
/
```

```
SQL> SELECT * FROM students;

STUDENT_ID
----------
STUDENT_NAME
------------------------------
      MARK GRADE
---------- ----------
         1
Aung Aung
        85

         2
Mg Mg
        70

STUDENT_ID
----------
STUDENT_NAME
------------------------------
      MARK GRADE
---------- ----------

         3
Su Su
        45

         4
Hla Hla

STUDENT_ID
----------
STUDENT_NAME
------------------------------
      MARK GRADE
---------- ----------
        30
```

/* Run Procedure */

```
SQL> BEGIN
    update_student_grade(1);
    update_student_grade(2);
    update_student_grade(3);
    update_student_grade(4);
END;
/
  2    3    4    5    6    7  Student grade updated successfully
Student grade updated successfully
Student grade updated successfully
Student grade updated successfully

PL/SQL procedure successfully completed.
```

/* Check Result */

```
SQL> SELECT * FROM students;

STUDENT_ID
----------
STUDENT_NAME
------------------------------
      MARK GRADE
---------- ----------
         1
Aung Aung
        85 A

         2
Mg Mg
        70 B

STUDENT_ID
----------
STUDENT_NAME
------------------------------
      MARK GRADE
---------- ----------

         3
Su Su
        45 C

         4
Hla Hla

STUDENT_ID
----------
STUDENT_NAME
------------------------------
      MARK GRADE
---------- ----------
        30 Fail
```

```
/* ============================================================
   9. FUNCTION Example
   Function to return grade by mark
   ============================================================ */
CREATE OR REPLACE FUNCTION get_grade
(
    p_mark IN NUMBER
)
RETURN VARCHAR2
AS
    v_grade VARCHAR2(10);
BEGIN
    IF p_mark >= 80 THEN
```

```
        v_grade := 'A';
    ELSIF p_mark >= 60 THEN
        v_grade := 'B';
    ELSIF p_mark >= 40 THEN
        v_grade := 'C';
    ELSE
        v_grade := 'Fail';
    END IF;

    RETURN v_grade;
END;
/

/* Run Function */
```

```
SQL> SQL> BEGIN
    DBMS_OUTPUT.PUT_LINE('Grade is: ' || get_grade(90));
    DBMS_OUTPUT.PUT_LINE('Grade is: ' || get_grade(55));
END;
  2    3    4    5  /
Grade is: A
Grade is: C

PL/SQL procedure successfully completed.
```

/* Use Function in SELECT */

```
SQL> SELECT
    student_id,
    student_name,
    mark,
    get_grade(mark) AS calculated_grade
FROM students;
  2    3    4    5    6
STUDENT_ID
----------
STUDENT_NAME
----------------------------------------
      MARK
----------
CALCULATED_GRADE
----------------------------------------
         1
Aung Aung
        85
A


STUDENT_ID
----------
STUDENT_NAME
----------------------------------------
      MARK
----------
CALCULATED_GRADE
----------------------------------------
         2
Mg Mg
        70
B
```

```
/* ============================================================
   10. EXCEPTION Example
   ============================================================ */
```

```
SQL> DECLARE
    v_student_name students.student_name%TYPE;
BEGIN
    SELECT student_name
    INTO v_student_name
    FROM students
    WHERE student_id = 100;

    DBMS_OUTPUT.PUT_LINE('Student Name: ' || v_student_name);

EXCEPTION
    WHEN NO_DATA_FOUND THEN
        DBMS_OUTPUT.PUT_LINE('No student found with this ID');

    WHEN TOO_MANY_ROWS THEN
        DBMS_OUTPUT.PUT_LINE('Query returned more than one row');

    WHEN OTHERS THEN
        DBMS_OUTPUT.PUT_LINE('Unexpected Error: ' || SQLERRM);
END;
/
  2    3    4    5    6    7    8    9   10   11   12   13   14   15   16   17
   18   19   20   21  No student found with this ID

PL/SQL procedure successfully completed.
```

```
/* ============================================================
   11. Complete Example: Loop + Procedure + Function
   Update all students grade using FOR LOOP
   ============================================================ */
```

```
SQL> BEGIN
    FOR rec IN
    (
        SELECT student_id
        FROM students
    )
    LOOP
        update_student_grade(rec.student_id);
    END LOOP;

    DBMS_OUTPUT.PUT_LINE('All student grades updated');
END;
/
  2    3    4    5    6    7    8    9   10   11   12   13
Student grade updated successfully
Student grade updated successfully
Student grade updated successfully
Student grade updated successfully
All student grades updated

PL/SQL procedure successfully completed.
```

/* Final Result */

```
SQL> SELECT * FROM students;

STUDENT_ID
----------
STUDENT_NAME
------------------------------
      MARK GRADE
---------- ----------
         1
Aung Aung
        85 A

         2
Mg Mg
        70 B

STUDENT_ID
----------
STUDENT_NAME
------------------------------
      MARK GRADE
---------- ----------

         3
Su Su
        45 C

         4
Hla Hla

STUDENT_ID
----------
STUDENT_NAME
------------------------------
      MARK GRADE
---------- ----------
        30 Fail
```